# CS23532-COMPUTER NETWORKS-LAB MANUAL

## Practical -1

## AIM: - Study of various Network commands used in Linux and Windows:

# Windows commands

C:\Users\Rajesh.B >hostname

Rajesh.B

C:\Users\Rajesh.B >ipconfig /all

Windows IP Configuration

Host Name . . . . . : Rajesh.B

Primary Dns Suffix . . . . . :

Node Type . . . . . : Mixed

IP Routing Enabled. . . . . : No

WINS Proxy Enabled. . . . . : No

Ethernet adapter Ethernet 2:

Connection-specific DNS Suffix . :

Description . . . . . : VirtualBox Host-Only Ethernet Adapter

Physical Address. . . . . : 0A-00-27-00-00-0E

DHCP Enabled. . . . . : No

Autoconfiguration Enabled . . . . : Yes

Link-local IPv6 Address . . . . . : fe80::aa4a:775:1de1:1757%14(Preferred)

IPv4 Address. . . . . : 192.168.56.1(Preferred)

Subnet Mask . . . . . : 255.255.255.0

Default Gateway . . . . . :

DHCPv6 IAID . . . . . : 118095911

DHCPv6 Client DUID. . . . . : 00-01-00-01-2D-79-FD-43-2C-58-B9-DE-5F-5F

NetBIOS over Tcpip. . . . . : Enabled

Wireless LAN adapter Local Area Connection* 1:

Media State . . . . . : Media disconnected

Connection-specific DNS Suffix . :

Description . . . . . : Microsoft Wi-Fi Direct Virtual Adapter

Physical Address. . . . . : F8-FE-5E-3A-70-56

DHCP Enabled. . . . . : Yes

Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Local Area Connection* 2:

Media State . . . . . : Media disconnected

Connection-specific DNS Suffix . :

Description . . . . . : Microsoft Wi-Fi Direct Virtual Adapter #2

Physical Address. . . . . : FA-FE-5E-3A-70-55

DHCP Enabled. . . . . : No

Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Wi-Fi:

Connection-specific DNS Suffix . :

Description . . . . . : Intel(R) Wi-Fi 6E AX211 160MHz

Physical Address. . . . . : F8-FE-5E-3A-70-55

DHCP Enabled. . . . . : Yes

Autoconfiguration Enabled . . . . : Yes

Link-local IPv6 Address . . . . . : fe80::b810:29c9:55e9:3a65%17(Preferred)

IPv4 Address. . . . . : 10.83.96.83(Preferred)

Subnet Mask . . . . . : 255.255.255.0

Lease Obtained. . . . . : 26 August 2026 14:38:14

Lease Expires . . . . . : 26 August 2026 15:38:11

Default Gateway . . . . . : 10.83.96.20

DHCP Server . . . . . : 10.83.96.20

DHCPv6 IAID . . . . . : 167312990

DHCPv6 Client DUID. . . . . : 00-01-00-01-2D-79-FD-43-2C-58-B9-DE-5F-5F

DNS Servers . . . . . : 10.83.96.20

NetBIOS over Tcpip. . . . . : Enabled

Ethernet adapter Bluetooth Network Connection:

Media State . . . . . : Media disconnected

Connection-specific DNS Suffix . :

Description . . . . . : Bluetooth Device (Personal Area Network)

Physical Address. . . . . : F8-FE-5E-3A-70-59

DHCP Enabled. . . . . : Yes

Autoconfiguration Enabled . . . . : Yes

Ethernet adapter Ethernet:

Media State . . . . . : Media disconnected

Connection-specific DNS Suffix . :

Description . . . . . : Realtek Gaming GbE Family Controller

Physical Address. . . . . : 2C-58-B9-DE-5F-5F

DHCP Enabled. . . . . : Yes

Autoconfiguration Enabled . . . . : Yes

C:\Users\Rajesh.B >arp -a

Interface: 192.168.56.1 --- 0xe

| Internet Address | Physical Address | Type |
|---|---|---|
| 192.168.56.255 | ff-ff-ff-ff-ff-ff | static |
| 224.0.0.22 | 01-00-5e-00-00-16 | static |
| 224.0.0.251 | 01-00-5e-00-00-fb | static |
| 224.0.0.252 | 01-00-5e-00-00-fc | static |
| 239.255.255.250 | 01-00-5e-7f-ff-fa | static |
| 255.255.255.255 | ff-ff-ff-ff-ff-ff | static |

Interface: 10.83.96.83 --- 0x11

| Internet Address | Physical Address | Type |
|---|---|---|
| 10.83.96.20 | 4e-ec-85-ec-ff-15 | dynamic |
| 10.83.96.255 | ff-ff-ff-ff-ff-ff | static |
| 224.0.0.22 | 01-00-5e-00-00-16 | static |
| 224.0.0.251 | 01-00-5e-00-00-fb | static |
| 224.0.0.252 | 01-00-5e-00-00-fc | static |
| 239.255.255.250 | 01-00-5e-7f-ff-fa | static |
| 255.255.255.255 | ff-ff-ff-ff-ff-ff | static |

C:\Users\Rajesh.B >netstat -r

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Interface List

14...0a 00 27 00 00 0e ......VirtualBox Host-Only Ethernet Adapter

11...f8 fe 5e 3a 70 56 ......Microsoft Wi-Fi Direct Virtual Adapter

4...fa fe 5e 3a 70 55 ......Microsoft Wi-Fi Direct Virtual Adapter #2

17...f8 fe 5e 3a 70 55 ......Intel(R) Wi-Fi 6E AX211 160MHz

19...f8 fe 5e 3a 70 59 ......Bluetooth Device (Personal Area Network)

2...2c 58 b9 de 5f 5f ......Realtek Gaming GbE Family Controller

1...........................Software Loopback Interface 1

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IPv4 Route Table

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Active Routes:

| Network Destination | Netmask | Gateway | Interface | Metric |
|---|---|---|---|---|
| 0.0.0.0 | 0.0.0.0 | 10.83.96.20 | 10.83.96.83 | 35 |
| 10.83.96.0 | 255.255.255.0 | On-link | 10.83.96.83 | 291 |
| 10.83.96.83 | 255.255.255.255 | On-link | 10.83.96.83 | 291 |
| 10.83.96.255 | 255.255.255.255 | On-link | 10.83.96.83 | 291 |
| 127.0.0.0 | 255.0.0.0 | On-link | 127.0.0.1 | 331 |
| 127.0.0.1 | 255.255.255.255 | On-link | 127.0.0.1 | 331 |
| 127.255.255.255 | 255.255.255.255 | On-link | 127.0.0.1 | 331 |
| 192.168.56.0 | 255.255.255.0 | On-link | 192.168.56.1 | 281 |
| 192.168.56.1 | 255.255.255.255 | On-link | 192.168.56.1 | 281 |
| 192.168.56.255 | 255.255.255.255 | On-link | 192.168.56.1 | 281 |
| 224.0.0.0 | 240.0.0.0 | On-link | 127.0.0.1 | 331 |
| 224.0.0.0 | 240.0.0.0 | On-link | 192.168.56.1 | 281 |
| 224.0.0.0 | 240.0.0.0 | On-link | 10.83.96.83 | 291 |
| 255.255.255.255 | 255.255.255.255 | On-link | 127.0.0.1 | 331 |
| 255.255.255.255 | 255.255.255.255 | On-link | 192.168.56.1 | 281 |

```
  255.255.255.255  255.255.255.255        On-link       10.83.96.83    291
===========================================================================
Persistent Routes:
  None

IPv6 Route Table
===========================================================================
Active Routes:
 If Metric Network Destination      Gateway
  1    331 ::1/128                  On-link
 14    281 fe80::/64                On-link
 17    291 fe80::/64                On-link
 14    281 fe80::aa4a:775:1de1:1757/128
                            On-link
 17    291 fe80::b810:29c9:55e9:3a65/128
                            On-link
  1    331 ff00::/8                 On-link
 14    281 ff00::/8                 On-link
 17    291 ff00::/8                 On-link
===========================================================================
Persistent Routes:
  None

C:\Users\Rajesh.B   >nslookup www.google.com
Server:  UnKnown
Address:  10.83.96.20

Non-authoritative answer:
```

Name: www.google.com

Addresses: 2001:4860:482d:7700::

2001:4860:4826:7700::

2001:4860:4828:7700::

2001:4860:482b:7700::

2001:4860:4827:7700::

2001:4860:4829:7700::

2001:4860:482a:7700::

2001:4860:482c:7700::

142.251.152.119

142.251.156.119

142.251.154.119

142.251.151.119

142.251.153.119

142.251.157.119

142.251.150.119

142.251.155.119

C:\Users\Rajesh.B >ping google.com

Pinging google.com [142.251.43.46] with 32 bytes of data: Reply

from 142.251.43.46: bytes=32 time=14ms TTL=114

Reply from 142.251.43.46: bytes=32 time=233ms TTL=114

Reply from 142.251.43.46: bytes=32 time=59ms TTL=114

Reply from 142.251.43.46: bytes=32 time=328ms TTL=114

Ping statistics for 142.251.43.46:

```
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 14ms, Maximum = 328ms, Average = 158ms

C:\Users\Rajesh.B   >route print
===========================================================================
Interface List
 14...0a 00 27 00 00 0e ......VirtualBox Host-Only Ethernet Adapter
 11...f8 fe 5e 3a 70 56 ......Microsoft Wi-Fi Direct Virtual Adapter
  4...fa fe 5e 3a 70 55 ......Microsoft Wi-Fi Direct Virtual Adapter #2
 17...f8 fe 5e 3a 70 55 ......Intel(R) Wi-Fi 6E AX211 160MHz
 19...f8 fe 5e 3a 70 59 ......Bluetooth Device (Personal Area Network)
  2...2c 58 b9 de 5f 5f ......Realtek Gaming GbE Family Controller
  1...........................Software Loopback Interface 1
===========================================================================

IPv4 Route Table
===========================================================================
Active Routes:
Network Destination        Netmask          Gateway       Interface  Metric
          0.0.0.0          0.0.0.0      10.83.96.20      10.83.96.83     35
       10.83.96.0    255.255.255.0         On-link       10.83.96.83    291
      10.83.96.83  255.255.255.255         On-link       10.83.96.83    291
     10.83.96.255  255.255.255.255         On-link       10.83.96.83    291
127.0.0.0        255.0.0.0         On-link         127.0.0.1    331
       127.0.0.1  255.255.255.255         On-link         127.0.0.1    331
  127.255.255.255  255.255.255.255         On-link         127.0.0.1    331
    192.168.56.0    255.255.255.0         On-link      192.168.56.1    281
```

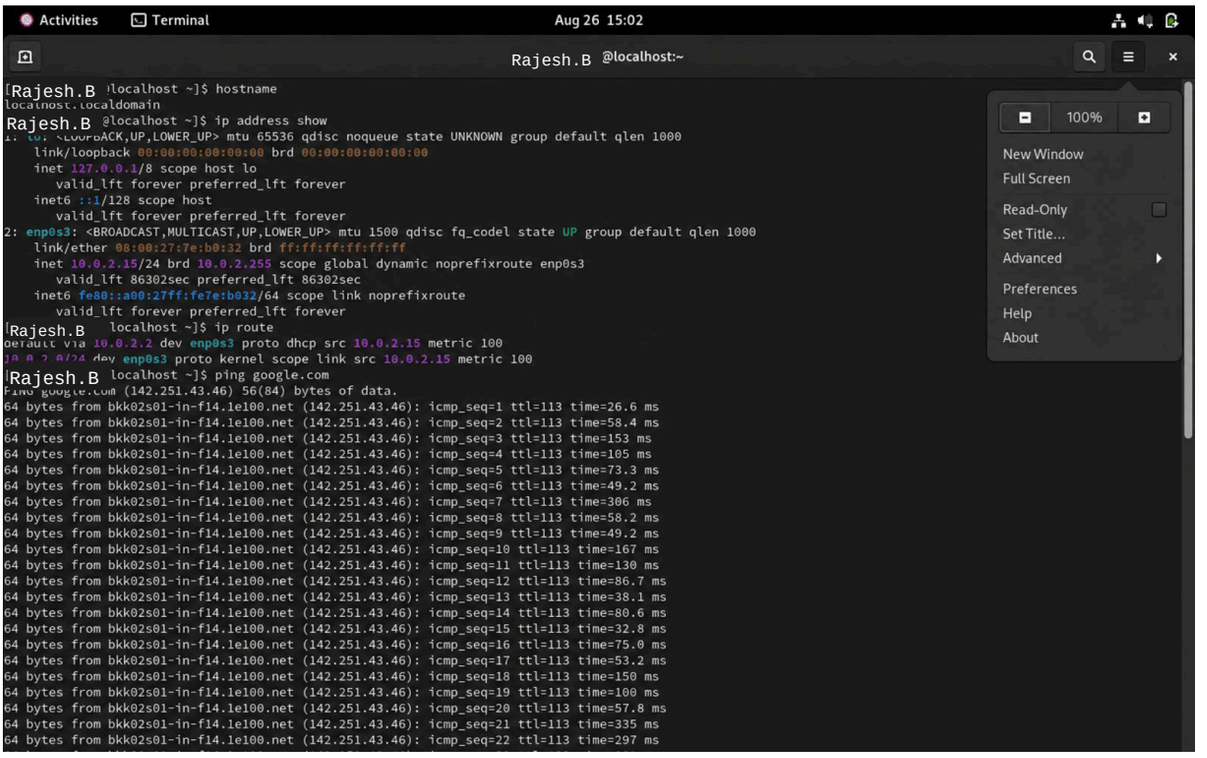
```
  192.168.56.1  255.255.255.255        On-link     192.168.56.1    281
192.168.56.255  255.255.255.255        On-link     192.168.56.1    281
    224.0.0.0       240.0.0.0        On-link        127.0.0.1    331
    224.0.0.0       240.0.0.0        On-link     192.168.56.1    281
    224.0.0.0       240.0.0.0        On-link      10.83.96.83    291
255.255.255.255  255.255.255.255        On-link        127.0.0.1    331
255.255.255.255  255.255.255.255        On-link     192.168.56.1    281
255.255.255.255  255.255.255.255        On-link      10.83.96.83    291
===========================================================================
Persistent Routes:
  None

IPv6 Route Table
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Active Routes:
 If Metric Network Destination      Gateway
  1    331 ::1/128                  On-link
 14    281 fe80::/64                On-link
 17    291 fe80::/64                On-link
 14    281 fe80::aa4a:775:1de1:1757/128
                                    On-link
 17    291 fe80::b810:29c9:55e9:3a65/128
                                    On-link
  1    331 ff00::/8                 On-link
 14    281 ff00::/8                 On-link
 17    291 ff00::/8                 On-link
===========================================================================
Persistent Routes:
```

None

C:\Users\Rajesh.B >tracert google.com

# Linux commands




**Student observation answers**

1. **Command used to check reachability:** ping
2. **Command used to find hops:** tracert in Windows / traceroute in Linux
3. **Command to display IP configuration:** ipconfig in Windows / ip address in Linux
4. **Command to display TCP port status:** netstat
5. **Command to modify IP configuration in Linux:** ip address / ip route